To find the ~~broad~~ last host address,
find the broadcast address

broadcast address = 11111111.11111111.11111111.111
00000
= 111111.11111111.11111111.11111
−1
= 192.168.100.159−1

Last host address = 192.168.100.158

d, Calculating Subnet Mask

Step Convert the subnet mark to binary

/27 = 27 bits are set to 1 and 5 bits are set to 0

So in binary = 11111111.11111111.11111111.11100000

Convert IP address to binary

192.168.100.154 = 11000000.10101000.01100100.10011010

Perform the AND operation

11111111.11111111.11111111.11100000
11000000.10101000.01100100.10011010
= 11000000.10101000.01100100.10010000

in decimal is 192.168.100.144 is network address

To find first host address

Use formula = network address + 1

= 192.168.100.144 + 1

first host address = 192.168.100.145